

QEMU 8.1 Released with New PipeWire Audio Backend, Many Improvements

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QEMU 8.1 open-source virtualization software has been released today as the first major update in the QEMU 8.0 series, bringing numerous new features and improvements. Highlights of QEMU 8.1 include a new PipeWire audio backend (`-audiodev pipewire`), along with better audio output and improved compatibility for audio devices. For the IBM System/390 architecture, QEMU 8.1 adds instruction emulation fixes for LDER, LCBB, LOCFHR, MXDB, MXDBR, EPSW, MDEB, MDEBR, MVCRL, LRA, CKSM, CLM, ICM, MC, STIDP, EXECUTE, and CLGEBR(A). PCIe support has been updated as well in QEMU 8.1, which introduces new QMP commands to inject CXL General Media events, DRAM events, and Memory Module events. Meanwhile, you can download QEMU 8.1 right now from the official website or install it from the stable software repositories of your favorite GNU/Linux distribution.

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Highlights of QEMU 8.1 include a new PipeWire audio backend (`-audiodev pipewire`), along with better audio output and improved compatibility for audio devices.

It also improves VFIO live migration support, which is no longer an experimental feature in this release, and updates the GTK graphical user interface to support multi-touch events.

For ARM, QEMU 8.1 brings AES acceleration for host processors, MTE support for guests in KVM virtual machines on hosts that supports the Memory Tagging Extension, emulation support for the Banana Pi BPI-M2 Ultra (bpim2u) single-board computer and the Cortex Neoverse-V1 (neoverse-v1) CPU, as well as support for new architectural feature support for FEAT_PAN3 (SCTLR_ELx.EPAN), FEAT_LSE2 (Large System Extensions v2), and experimental support for FEAT_RME (Realm Management Extensions).

For RISC-V, the new QEMU release also brings AES acceleration for host processors, along with ISA extension support for BF16/Zfa, disassembly support for Zcm, Zinx, XVentanaCondOps, and Xthead, CPU emulation support for Veyron V1, as well as numerous KVM/emulation fixes and enhancements.

PowerPC support has been improved as well with AES acceleration for host processors, as well as TCG SMT support, allowing pseries and powernv to run with up to 8 threads per core, emulation support for the Power9 DD2.2 CPU model, and perf sampling support for POWER CPUs.

For the IBM System/390 architecture, QEMU 8.1 adds instruction emulation fixes for LDER, LCBB, LOCFHR, MXDB, MXDBR, EPSW, MDEB, MDEBR, MVCRL, LRA, CKSM, CLM, ICM, MC, STIDP, EXECUTE, and CLGEBR(A).

For SPARC, there's improved performance for target/sparc with support for `tcg_gen_lookup_and_goto_ptr()`, and for TriCore there's emulation support for TC37x CPU that supports ISA v1.6.2 instructions and instruction emulation of POPCNT.W, LHA, CRC32L.W, CRC32.B, SHUFFLE, SYSCALL, and DISABLE.

Last but not least, x86 gets CPU model support for GraniteRapids, Hexagon gets gdbstub support for HVX and new instruction support for v68/v73 scalar, and v68/v69 HVX, and MIPS gets emulation support for Ingenic XBurstR1 and XBurstR2 CPUs, and MXU instructions.

PCIe support has been updated as well in QEMU 8.1, which introduces new QMP commands to inject CXL General Media events, DRAM events, and Memory Module events.

For more details on the changes implemented in this release, check out the full changelog. Meanwhile, you can download QEMU 8.1 right now from the official website or install it from the stable software repositories of your favorite GNU/Linux distribution.

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<https://9to5linux.com/qemu-8-1-released-with-new-pipewire-audio-backend-many-improvements>